

Exact Exchange in Orbital-Free Density Functional Theory

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In this work, we revisit the Levy–Perdew–Sahni (LPS) formalism for the square root of the electron density, starting from a reference system of non-interacting boson-like fermions. This perspective preserves the conventional form of the theory while analytically demonstrating that the Fermi–Amaldi (FA) functional is the exact orbital-free exchange energy. This advancement resolves the lack of the exact exchange density functional in orbital-free density functional theory (OF-DFT) opening new avenues for constructing orbital-free functionals beyond the generalized gradient approximation (GGA). The numerical study of the inclusion of the exact exchange in the LPS scheme is provided.

I. INTRODUCTION

The most popular practical method in density functional theory, Kohn–Sham DFT, is exact in principle and offers a favorable balance of accuracy and computational cost [1–3]. The implementation of KS-DFT scales nearly as $\mathcal{O}(N^3)$ limiting the applicability of KS-DFT to large systems, such as periodic structures and proteins [4].

The Levy–Perdew–Sahni formulation of orbital-free density functional theory offers an alternative that facilitates near-linear scaling within standard KS codes [4–6]. It models the electronic system as N non-interacting bosons governed by a single Schrödinger-like equation for the square root of the electron density $n(\mathbf{r})$

$$\left\{ -\frac{1}{2}\nabla^2 + v_{\text{LPS}}([n]; \mathbf{r}) \right\} \sqrt{n(\mathbf{r})} = \mu \sqrt{n(\mathbf{r})}, \quad (1)$$

where μ is the chemical potential [5]. The effective LPS potential enforces the correct physical density by augmenting the conventional KS potential with a Pauli potential term

$$v_{\text{LPS}}([n]; \mathbf{r}) = v_{\text{KS}}([n]; \mathbf{r}) + \frac{\delta T_{\text{P}}[n]}{\delta n(\mathbf{r})}, \quad (2)$$

$$v_{\text{KS}}([n]; \mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + \frac{\delta E_{\text{xc}}[n]}{\delta n(\mathbf{r})} \quad (3)$$

Here, $T_{\text{P}}[n]$ is the Pauli kinetic energy functional, $v_{\text{ext}}(\mathbf{r})$ is the external nuclear potential, and $E_{\text{xc}}[n]$ is the exchange-correlation functional.

The widespread application of the LPS OF-DFT necessitates accurate orbital-free approximations for both the kinetic, $T_{\text{P}}[n]$, and exchange-correlation, $E_{\text{xc}}[n]$, components of the energy. While significant progress has been made for the first one, ranging from local and semi-local to advanced machine-learned approximations, the orbital-free exchange-correlation functionals are limited to lower rungs of Jacob’s ladder (GGA and meta-GGA). The transfer of the vast library of KS-DFT approximations, including highly-accurate hybrid approximations, to the orbital-free framework is currently hindered by the absence of an orbital-free exact exchange functional [4, 6–9].

II. DERIVATION

In this section we derive the exact orbital-free exchange functional. We reexamine the original LPS formalism [5] from the viewpoint of a reference system of non-interacting boson-like fermions. This perspective preserves the conventional structure of the theory while yielding the exact orbital-free exchange functional.

In their original work, Levy, Perdew, and Sahni considered a system of N non-interacting bosons with the interacting energy functional $E[n]$ given by

$$E[n] = T_{\text{W}}[n] + T_{\text{P}}[n] + V_{\text{ext}}[n] + J[n] + E_{\text{xc}}[n]. \quad (4)$$

The kinetic energy of the non-interacting bosons is given by the von Weizsäcker kinetic energy functional [10]

$$T_{\text{W}}[n] = \int \sqrt{n(\mathbf{r})} \left(-\frac{1}{2}\nabla^2 \right) \sqrt{n(\mathbf{r})} d\mathbf{r}. \quad (5)$$

The Pauli term, $T_{\text{P}}[n]$, is defined as a difference between the KS non-interacting kinetic energy functional [3] and the von Weizsäcker term

$$T_{\text{P}}[n] = T_{\text{s}}[n] - T_{\text{W}}[n]. \quad (6)$$

Subtracting $T_{\text{W}}[n]$ enables solving Eq. (1) for kinetic energy functionals lacking an explicit Laplacian operator, e.g., the Thomas–Fermi (TF) functional [11, 12], while preserving the KS eigenvalue problem form [5, 13].

The remaining terms of Eq. (4) include the external potential energy, $V_{\text{ext}}[n]$, and the classical Coulomb self-repulsion energy, $J[n]$

$$J[n] = \frac{1}{2} \int \int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}d\mathbf{r}'. \quad (7)$$

While the original LPS formalism was derived for bosons, the authors noted its potential generalization to other types of particles. Here, we formally extend this to a fictitious system of non-interacting boson-like fermions with spin degeneracy $M = 2s + 1$ equal to the particle number to derive the exact exchange.

The non-interacting wave function for such a system is a Slater determinant of N orthonormal spin-orbitals

$\{\psi_i(\mathbf{x})\}$, each formed as a product of a spatial function $\sqrt{n(\mathbf{r})/N}$ and an orthogonal spin function $\{\sigma_i(\omega)\}$. Since every particle occupies a distinct spin state, the Pauli exclusion principle permits them to share the same spatial orbital. Consequently, the antisymmetry is confined entirely to the spin part of the wave function, and the it factors as

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \frac{\Phi_{\text{HP}}(\mathbf{r}_1, \dots, \mathbf{r}_N)}{\sqrt{N!}} \det\{\sigma_i(\omega_j)\}, \quad (8)$$

where the spatial part Φ_{HP} is the Hartree product [14]. Evaluating the non-interacting kinetic energy as an expectation value of the wave function of Eq. (8) yields the von Weizsäcker kinetic energy functional $T_{\text{W}}[n]$. Thus, the LPS energy functional (4) [5] can be fully recovered from here while physically reinterpreting the reference system as non-interacting boson-like fermions.

Explicit choice of the wave function enables applying the KS exact exchange definition to the LPS framework

$$E_{\text{x}}^{\text{exact}}[n] = \langle \Psi_{\text{min}} | \hat{V}_{\text{ee}} | \Psi_{\text{min}} \rangle - J[n], \quad (9)$$

$$\hat{V}_{\text{ee}} = \sum_{i < j}^N \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad (10)$$

where $\hat{V}_{\text{ee}}$ is the two-electron operator, and Ψ_{min} is the energy-minimizing wave function of the form given in Eq. (8). Since $\hat{V}_{\text{ee}}$ is spin-independent, it applies directly to our boson-like fermion wave function. Substituting Eq. (8) into Eq. (9) yields the Fermi–Amaldi functional [15]

$$E_{\text{x}}^{\text{exact}}[n] \equiv E_{\text{x}}^{\text{FA}}[n] = -\frac{1}{N} J[n]. \quad (11)$$

This result aligns with our previous findings [16, 17] and confirms that the Fermi–Amaldi functional constitutes the exact exchange energy for any system where particles share a single spatial orbital, independent of spin statistics.

III. NUMERICAL ILLUSTRATIONS

In this section, we assess the impact of the FA functional on atomic energies, dissociation curves, and electron densities. To achieve this, we numerically solve Eq. (1) using a modified KS-DFT code. The effective potential is constructed according to Eqs. (2) and (3) by setting $E_{\text{xc}}[n] = E_{\text{x}}^{\text{FA}}[n]$. This yields the explicit FA potential

$$v_{\text{x}}^{\text{FA}}([n]; \mathbf{r}) = -\frac{1}{N} \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (12)$$

where N is treated as a constant parameter [18–24]. For assessment purposes, the FA functional is tested against the local LDA [25, 26] and the semi-local (GGA) PBE [27, 28] exchange functionals. The specific choices of the Pauli term are detailed in the following sections.

A. Implementation Details

To implement the LPS scheme of Eq. (1), we adapted the open-source Psi4NumPy code [29, 30] by constructing the density matrix exclusively from the lowest-energy eigenfunction normalized to N [5]. The current work utilizes the spin-polarized formalism: the α and β density matrices are generated from the lowest-energy eigenfunctions normalized to N_{α} and N_{β} , respectively.

A Core Hamiltonian initial guess was employed, and all molecular integrals were computed analytically using a Gaussian-type basis set [14]. The approximate functionals $T_{\text{P}}[n]$ and $E_{\text{x}}[n]$ were evaluated numerically using a Lebedev–Treutler (999, 974) quadrature grid. In difficult convergence cases, the electron density was calculated at smaller, (350, 74), grid and was used as an initial guess for the larger grid calculations.

To obtain the self-consistent solutions, we used a combination of the direct inversion in the iterative subspace (DIIS) procedure [31] with Hartree damping [32] and Level-Shifting [33] techniques. The maximum length of the DIIS vector was varied from 3 to 9 depending on the system. All eigenvalues except the lowest one were shifted up by 0.1–0.9 at the beginning of the SCF cycle returning it back to 0.0 closer to the convergence. The damping factor was varied in the range 0.1–0.999. The convergence criteria for both the total energy and RMSD of the density matrix was set to 1.0E-5 Hartree and unitless change, respectively.

To validate our implementation, we reproduced the reference H–Ne results of Chan *et al.* [34] and Karasiev *et al.* [35]. For open-shell atoms, this required adapting our code to match the spin-restricted approach used in those studies.

B. Self-Consistent Atomic Energies

Self-consistent calculations were performed using the Thomas–Fermi- λ -Weizsäcker (TF λ W) [34–37] Pauli term

$$T_{\text{P}}^{\text{TF}\lambda\text{W}}[n] = T_{\text{TF}} + (\lambda - 1)T_{\text{W}}, \quad (13)$$

where $T_{\text{TF}}[n] = C_{\text{F}} \int n^{5/3}(\mathbf{r}) d\mathbf{r}$, and $C_{\text{F}} = 3/10(3\pi^2)^{2/3}$. Specific λ values were taken from the literature (see [34, 35] and references therein). Despite the known deficiencies [4, 38], the TF λ W model was selected because it is possible to bring it to the self-consistent convergence using modified KS code [34, 35].

Table I presents the mean absolute errors MAE = $(\sum_a \Delta E_a)/N_A$, where $\Delta E_a = |E_a^{\text{LPS}} - E_a^{\text{UHF}}|$, across $N_A = 10$ atoms, in a.u., and the relative mean absolute errors rMAE = $(-\sum_a \Delta E_a/E_a^{\text{UHF}})/N_A \times 100\%$ expressed as a percentage of the UHF/UGBS energy.

To maintain the spherical symmetry of the atoms and assist the convergence, a UGBS basis set was contracted to 31 s -type Gaussians, $\exp(-\alpha r^2)$, with exponents α ranging from $1.14 \times 10^7 \dots 2.00 \times 10^{-2}$, for all calcula-

TABLE I. H–Ne atomic SCF energy errors (relative to UHF/UGBS) for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TFW kinetic energy functionals. The contracted UGBS31 basis set was used for all atoms (see Sec. III B).

λ	1/9	1/6	0.186	0.2	1/3	1.0
Mean Absolute Error (a.u.)						
LDA	4.38	1.58	0.76	0.24	4.23	15.74
PBE	4.95	2.11	1.26	0.69	3.78	15.44
FA	2.91	0.23	0.65	1.19	5.48	16.67
Relative Mean Absolute Error (%)						
LDA	9.67	3.72	2.09	1.72	12.24	40.24
PBE	12.54	5.10	3.03	1.88	10.07	38.86
FA	8.00	1.14	1.66	2.71	13.41	41.00

tions involving the TFW model. This basis set will be referred to as UGBS31.

In alignment with other authors [34, 35], Table I shows that the optimal von Weizsäcker mixing parameter for the PBE and LDA pure exchange functionals is $\lambda = 1/5$. As for the FA functional, the optimal value is $\lambda = 1/6$, with the second best being $\lambda = 0.186$. This result indicates that the FA functional can outperform both LDA and PBE exchange functionals when paired with an appropriate kinetic energy density functional.

C. Atomic Densities

Self-consistent radial distribution functions for the Ne atom are presented in Fig. 1. The optimal TFW kinetic functionals were employed for each exchange functional. All calculations used the modified UGBS31 basis set described in the previous section.

The $\text{TF}_{\frac{1}{6}}\text{W}$ FA density decays more rapidly than either the $\text{TF}_{\frac{1}{5}}\text{W}$ LDA or $\text{TF}_{\frac{1}{5}}\text{W}$ PBE densities. As expected, the choice of exchange functional does not fundamentally alter the density structure, i.e., the FA exchange does not recover atomic shell structure, which remains absent due to the monotonic nature of the TFW kinetic model. The more rapid asymptotic decay observed in the FA density is consistent with established results from statistical atomic theory [39].

D. Dissociation Profiles

Figure II shows the H_2 dissociation curves computed using the same models as in the previous section for density plotting. These calculations employed the UGBSP basis set, a modified UGBS containing 13 s - and 4 p -functions ($\alpha_s \in [9.34 \times 10^2, 7.67 \times 10^{-2}]$ and $\alpha_p \in [5.76 \times 10^{-1}, 3.92 \times 10^{-2}]$) for each hydrogen. An RHF/UGBSP curve is included as the reference. Although spin-polarized functionals were used, our LPS implementation lacks a symmetry-breaking mechanism. As such, it

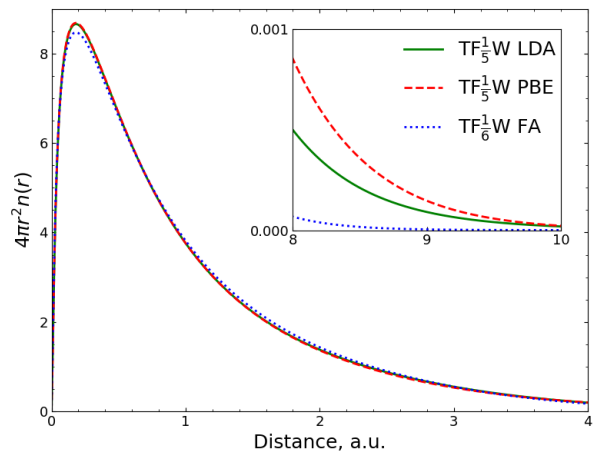


FIG. 1. Ne SCF radial distribution function $4\pi r^2 n(r)$ for Fermi–Amaldi, LDA, and PBE exchange functionals combined with $\text{TF}_{\frac{1}{6}}\text{W}$ and $\text{TF}_{\frac{1}{5}}\text{W}$ kinetic energy functionals, respectively. The contracted UGBS31 basis set was used (see Sec. III B).

converges to the spin-symmetric solution, making RHF the appropriate spin-symmetric reference rather than the broken-symmetry UHF solution.

As shown in Figure II, the FA functional exhibits better long-range behavior and a distinct potential well unlike both the LDA and PBE functionals. The FA functional also yields an equilibrium distance, r_e , significantly closer to the RHF reference, e.g., 2.00 a.u. against 3.00 and 3.25 a.u. for LDA and PBE, respectively. The FA binding energy estimate, $E(4 \text{ a.u.}) - E(r_e) = 0.0134 \text{ a.u.}$, is closer to the RHF reference (0.2191 a.u.) compared to 0.0028 and 0.0007 a.u. for LDA and PBE, respectively.

TABLE II. H–Ne atomic post-SCF energy errors (relative to UHF/UGBS) for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TF λ W, TF, Weizsäcker, and Tran–Wesołowski (TW) kinetic energy functionals. SCF densities were obtained at the UHF/UGBS level of theory.

λ	0.0	1/9	1/6	0.186	0.2	1/3	1.0 (no TF)	TW
Mean Absolute Error (a.u.)								
LDA	3.47	0.37	2.29	2.95	3.44	8.04	9.47	0.52
PBE	3.97	0.21	1.78	2.45	2.93	7.53	9.93	0.07
FA	1.92	1.91	3.83	4.50	4.98	9.58	7.90	2.06
Relative Mean Absolute Error (%)								
LDA	6.78	3.02	7.86	9.54	10.76	22.36	13.17	2.97
PBE	9.09	0.93	5.41	7.09	8.31	19.91	13.06	0.60
FA	5.38	4.29	9.12	10.80	12.02	23.62	9.35	4.24

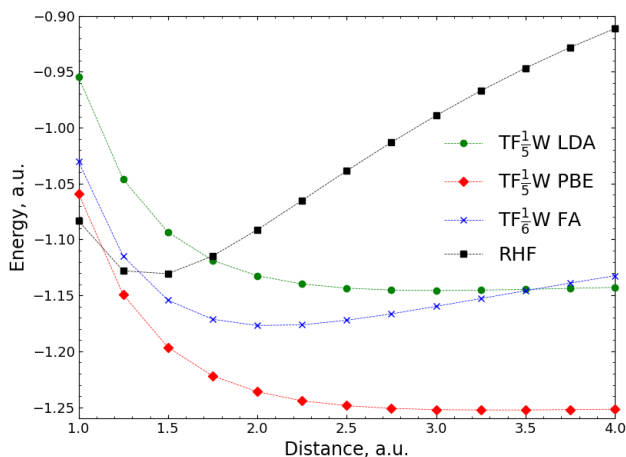


FIG. 2. H₂ SCF potential energy surface for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TF $\frac{1}{6}$ W and TF $\frac{1}{5}$ W kinetic energy functionals, respectively. Calculations were performed with the UGBSP basis set (see Sec. III D). Self-consistent calculations were performed only at the marked points. The connecting curves are interpolations.

E. Post-SCF Atomic Energies

The FA functional’s promising energetic performance on one hand with the absence of shell structure in the self-consistent densities on another prompted us to evaluate its accuracy in post-SCF calculations. The set of self-consistent UHF/UGBS atomic densities for H–Ne was used to compute post-SCF atomic energies with the FA, LDA, and PBE exchange functionals.

The range of kinetic energy density functionals employed in this section includes the TF λ W functionals from Table I as well as the pure TF [11, 12], the pure Weizsäcker [10], and the Tran–Wesołowski (TW) [40] functionals. The latter was chosen because it shares

the functional form with PBE [27, 28, 41, 42] and Becke86A [43] exchange functionals making it a good pair to the PBE exchange. The Pauli term for the TW kinetic functional is given by

$$T_P^{TW}[n] = C_F \int n^{5/3}(\mathbf{r}) F_{TW}(s) - T_W[n], \quad (14)$$

where $s = |\nabla n|/(2k_F n)$ is the reduced density gradient, $k_F = (3\pi^2 n)^{1/3}$ is the Fermi wave vector, and $F_{TW}(s)$ is the Tran–Wesołowski enhancement factor

$$F_{TW}(s) = 1 + \kappa - \frac{\kappa}{1 + \frac{\mu}{\kappa} s^2}, \quad (15)$$

with empirical parameters $\kappa = 0.8209$ and $\mu = 0.2335$ fitted to He and Ne atoms [40].

Table II summarizes the mean absolute errors (MAE) and relative mean absolute errors (rMAE) for each exchange-kinetic combination. Overall, the performance of the approximate exchange functionals strongly depends on the choice of the kinetic functional. The FA functional yields consistently higher errors than both LDA and PBE for all tested kinetic functionals except the pure TF functional. When paired with the TF λ W model, all exchange functionals achieve their lowest MAE and rMAE at $\lambda = 1/9$ which corresponds to the second-order gradient expansion of the TF functional. As expected, the PBE functional performs best when paired with the Tran–Wesołowski kinetic approximation.

IV. CONCLUSION

In this work, we have reinterpreted the Levy–Perdew–Sahni (LPS) scheme to demonstrate that the Fermi–Amaldi functional is the exact exchange energy within the orbital-free density functional theory. By utilizing a reference system of non-interacting boson-like fermions, where all particles share a single spatial orbital, we showed that the conventional LPS energy functional [5] is fully recovered and the exact exchange is easily defined.

The Fermi–Amaldi functional exhibits several highly desirable properties for an exchange functional. It is self-interaction free for one-orbital systems [16, 44], possesses the correct long-range asymptotic behavior [44], and exhibits the necessary derivative discontinuity at integer electron numbers N [18]. Furthermore, unlike the ex-

act Kohn–Sham exchange potential, the Fermi–Amaldi potential is strictly multiplicative. Recently, we also extended the FA functional to fractional electron number [18]. Collectively, these features identify the FA functional as a promising building block for future developments in OF-DFT.

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